



6 Model Selection and Diagnostics

In the previous tutorial, we discuss how to conduct inference on the parameters of the $\text{ARMA}(p, q)$ model given the order p and q . In this tutorial, we will discuss how to select a suitable value of p and q .

6.1 Graphical Method (For stationary AR or MA model)

Noticing that for $\text{MA}(q)$ model, i.e., $Y_t = Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$, we have

$$\gamma(k) = \begin{cases} \sigma^2 \sum_{j=0}^{q-|k|} \theta_j \theta_{j+|k|} & , \text{ if } |k| \leq q \\ 0 & , \text{ if } |k| > q \end{cases}$$

where $\theta_0 := 1$ for convention. It follows that ACVF $\gamma(\cdot)$ and hence the ACF plot will show a sharp cut-off at lag q . For example, see the following figure

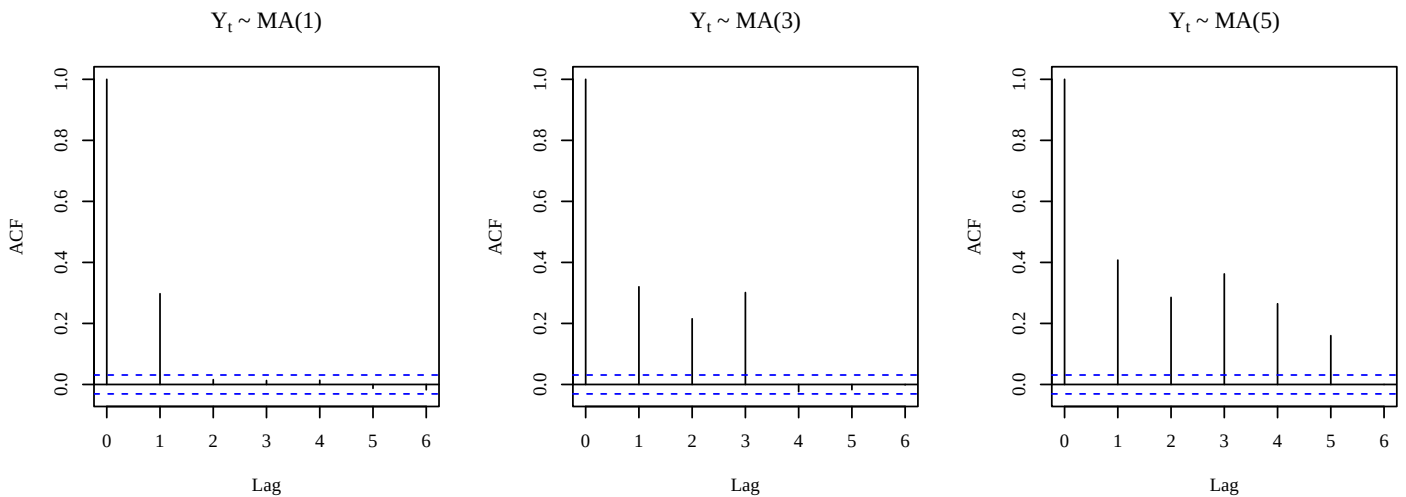


Figure 1: Examples of ACF plot of $\text{MA}(q)$ models

Recall from Tutorial 02 that $\sqrt{n}r_k \xrightarrow{d} N(0, 1)$ under the assumption that $\gamma(k) = 0$. Hence for large n , the confidence interval for r_k is given by

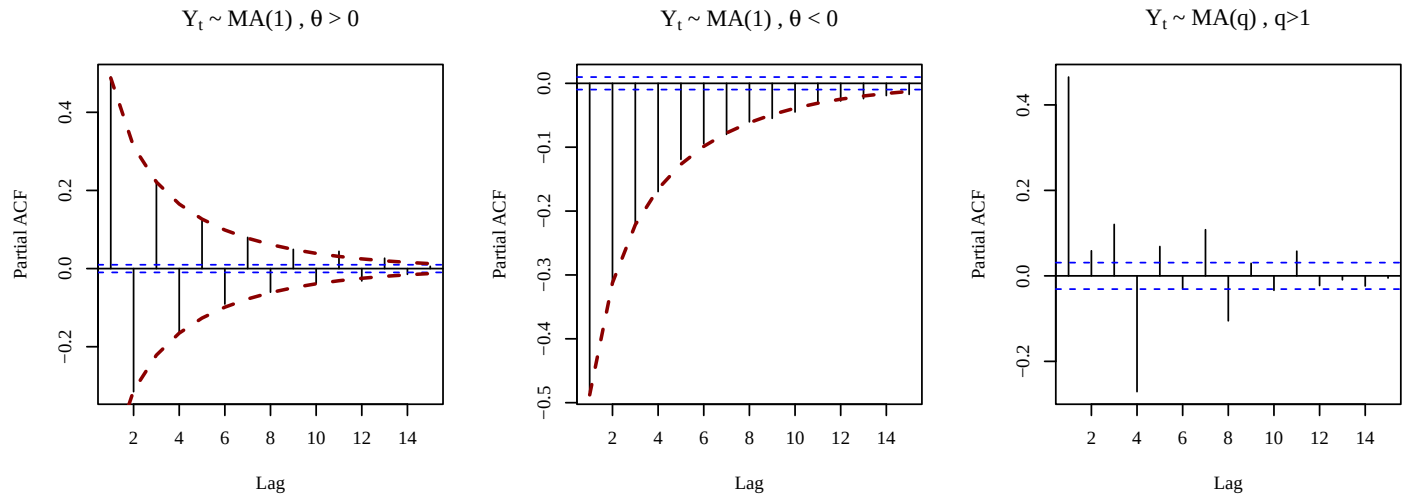
$$\hat{I} = \left[-\frac{z_{1-\alpha/2}}{\sqrt{n}}, \frac{z_{1-\alpha/2}}{\sqrt{n}} \right],$$

where $z_{1-\alpha/2}$ is the $100(1 - \alpha/2)\%$ quantile of standard normal distribution. The blue line in figure 1 refers to the confidence interval $\hat{I}$ with $\alpha = 0.05$. Therefore, we consider the value of ACF to be significantly different from 0 if it lies out of $\hat{I}$.

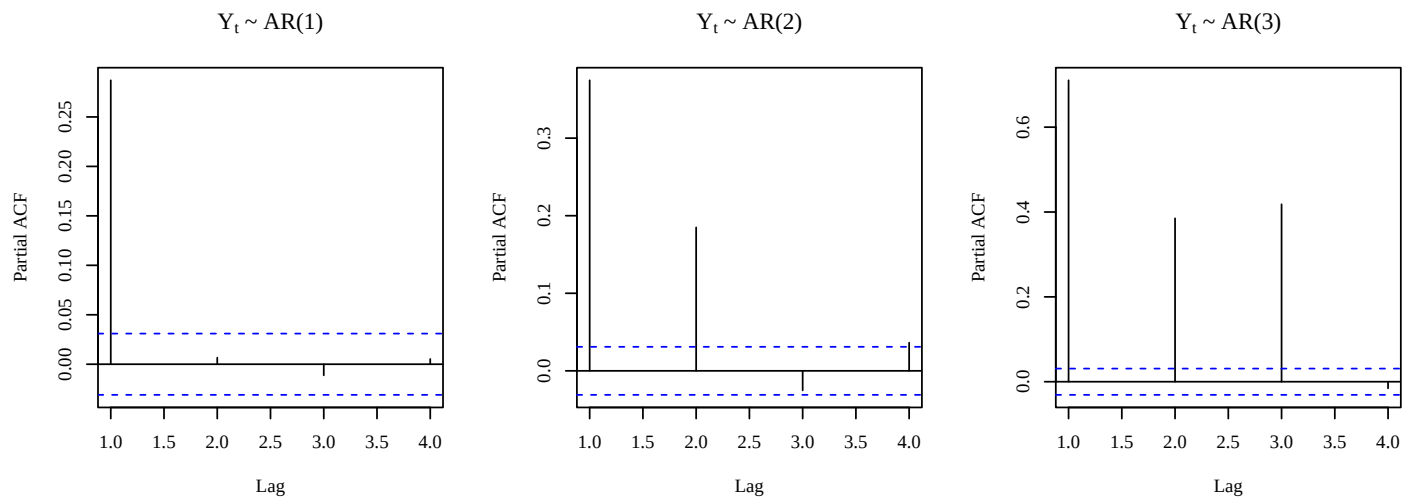
For the general $\text{MA}(q)$ model, it does not show a special pattern in the PACF plot. As a special case, for $\{Y_t\} \sim \text{MA}(1)$, we have

$$\phi_{kk} = -\frac{(-\theta)^k(1 - \theta^2)}{1 - \theta^{2(k+1)}},$$

which shows roughly exponential decay (in terms of magnitude). See the following figure

Figure 2: PACF plot of MA(1) and general MA(q) model

For the AR(p) model: $Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$, we have $\phi_{k,k} = 0$ whenever $k > p$, i.e., the PACF shows a sharp cut-off at lag p . For example, see

Figure 3: Examples of PACF plot of AR(p) models

For the general AR(p) model, it does not show a special pattern in the ACF plot. As a special case, for $\{Y_t\} \sim \text{AR}(1)$, we have

$$\rho(k) = \phi^{|k|}$$

and hence the ACF plot for AR(1) model shows an exponential decay in magnitude. See figure 4.

Remark 1. MA(q) must be stationary. However, the AR(p) model is not stationary in general. The graphical method can only help to specify a stationary AR or MA model.

The following is a short summary of the graphical model

Plot of	MA(q)	AR(p)	MA(1)	AR(1)
ACF $\rho(k)$	Cut-off at lag q	NA	Cut-off at lag 1	Exponential decay
PACF ϕ_{kk}	NA	Cut-off at lag p	Exponential Decay	Cut-off at lag 1

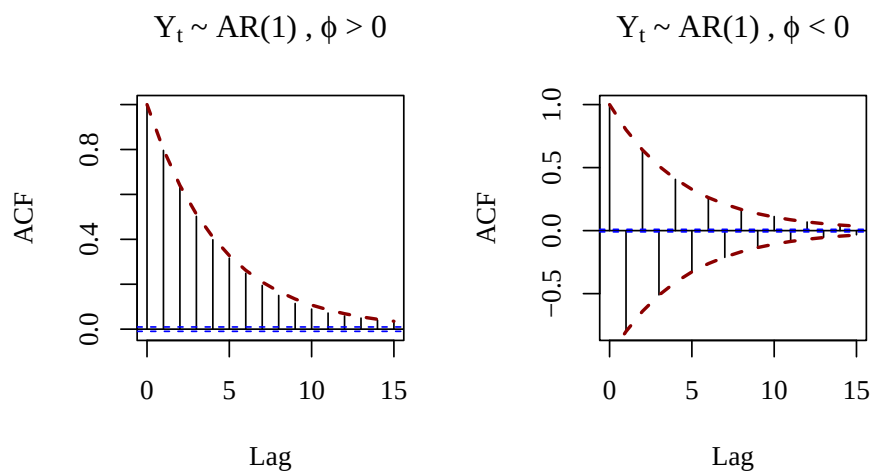
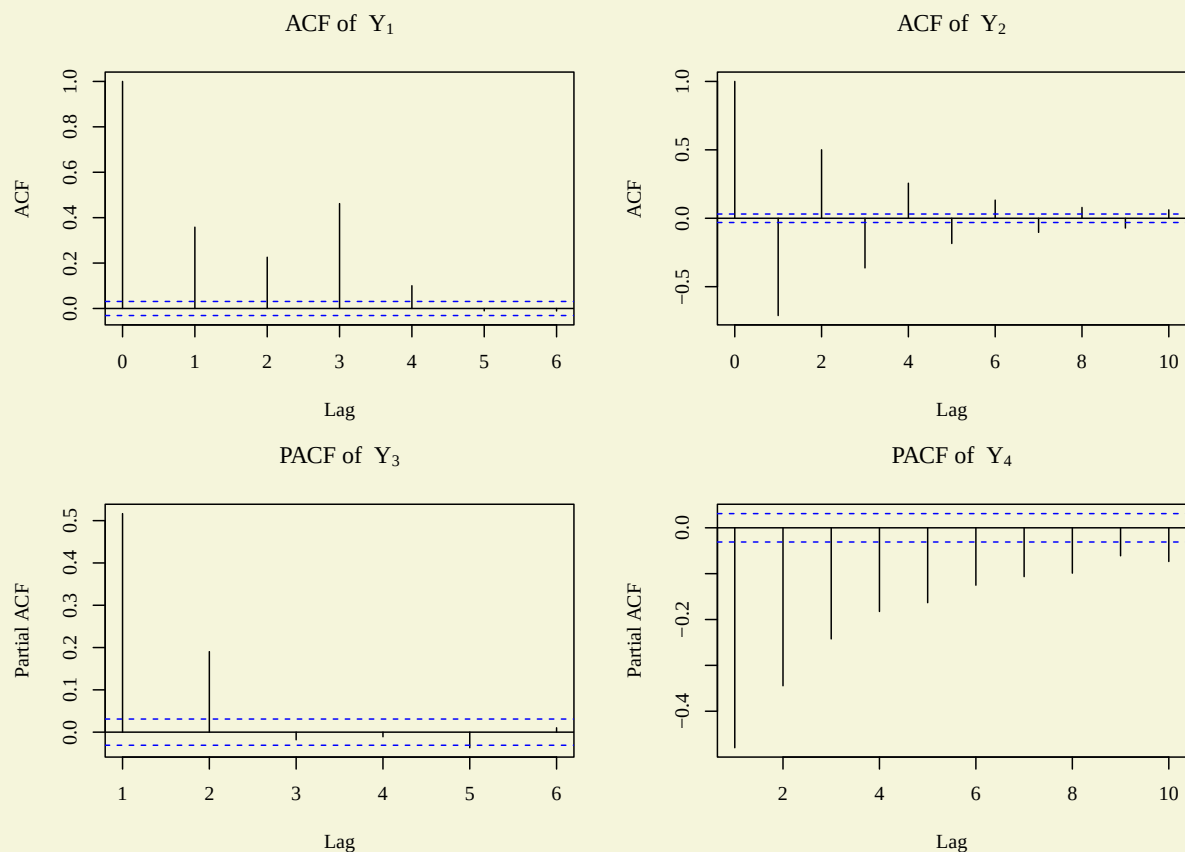


Figure 4: ACF plot of AR(1) models

(☆☆☆) Identification of Order (Graphical Method)

Exercise 1. Identify the following AR or MA model and specify the order of the model.



Attempt

6.2 Information Criteria (General Approach)

Notice that the graphical method is hard to determine the order of the general ARMA(p, q) model and is a little bit ambiguous. Instead, we can decide on some criteria function for proceeding with rigorous model selection. In this subsection, we consider ARMA(p, q) model with

- $\hat{\beta} = (\hat{\phi}_1, \dots, \hat{\phi}_p, \hat{\theta}_1, \dots, \hat{\theta}_q)$ and $\hat{\sigma}^2$ as the MLE of the model given $Y_1, \dots, Y_n$.
- $S_Y(\hat{\beta}) = \sum_{t=1}^n \hat{Z}_t^2$, where $\hat{Z}_t = Y_t - \hat{\phi}_1 Y_{t-1} - \dots - \hat{\phi}_p Y_{t-p} - \hat{\theta}_1 Z_{t-1} - \dots - \hat{\theta}_q Z_{t-q}$.
- $L(\hat{\beta}, \hat{\sigma}^2) = (2\pi\hat{\sigma}^2)^{-n/2} \exp\{-S_Y(\hat{\beta})/(2\hat{\sigma}^2)\}$, as the likelihood function of ARMA(p, q) model.

Information Criteria

Definition 1. The following are definitions of some common criteria function

1. **AIC (Akaike's Information Criterion)**

$$-2 \log L \left(\hat{\beta}, \frac{S_Y(\hat{\beta})}{n} \right) + 2(p + q + 1),$$

2. **AICC (AIC corrected)**

$$-2 \log L \left(\hat{\beta}, \frac{S_Y(\hat{\beta})}{n} \right) + \frac{2(p + q + 1)n}{n - p - q - 2},$$

3. **BIC (Bayesian Information Criterion)**

$$(n - p - q) \log \left(\frac{n\hat{\sigma}^2}{n - p - 1} \right) + n(1 + \log \sqrt{2\pi}) + (p + q) \log \left(\frac{\sum_{i=1}^n Y_i^2 - n\hat{\sigma}^2}{p + q} \right).$$

4. **FPE (Final Prediction Error)** [For AR models only]

$$\left(\frac{n + p}{n - p} \right) \hat{\sigma}^2.$$

❗ **Remark 2.** Given a criteria function f , we should always choose the model which gives a lower value, i.e., given models $\mathcal{F}_1$ and $\mathcal{F}_2$, if $f(\mathcal{F}_1) < f(\mathcal{F}_2)$, we prefer $\mathcal{F}_1$ more.

❗ **Remark 3.** Different criteria functions are developed to achieve different goals.

1. **AIC** estimates the Expected Predictive Log-likelihood $E\{\log f(y_{n+1}|\hat{\beta})|y_1, \dots, y_n\}$, i.e. AIC is useful if one wishes to find a model which gives an accurate prediction.
2. For a small sample size n , it is likely that AIC will select a model with too many parameters, i.e., overfitted model. **AICC** is a corrected version of AIC so that it works well for small n .
3. **BIC** estimates the Marginal Log-likelihood $\log f(y_1, \dots, y_n|\hat{\beta})$, i.e. the intrinsic of the model. As $n \rightarrow \infty$, BIC always chooses the correct model. (Unsatisfactory performance for small n)
4. **FPE** estimates $E\{(\hat{\phi} - \phi)^T(\hat{\phi} - \phi)\} = \sum_{k=1}^p \text{MSE}(\hat{\phi}_k)$, i.e., sum of MSE of the parameter estimate. Notice that it is a valid measure of goodness of fit only for the AR model.

There are some useful R-commands in model fitting

1. `arma.sim(n, model=list( ar=c(phi1,..., phiq), ma=c(theta1,..., thetap) ))` is used for generation of $(Y_1, \dots, Y_n) \sim \text{ARMA}(p, q)$.
2. `arma(Y, order=c(p,d,q))` fit the vector Y to the $\text{ARIMA}(p, d, q)$ model.

(☆☆☆) Order Selection by Information Criteria

 **Exercise 2.** Read the attached R-code and answer the following:

- (a) Write down the order of the true model and the models fitted. (refer to lines 1 and 2)
- (b) Calculate the AIC and BIC of the models. Which model should you choose based on different criteria?

```
Y = arma.sim(405, model=list( ar=c(0.3,0.4,0.2), ma=c(0.2,0.1,0.6)))
Model_1 = arma(Y, order=c(2,0,3)) ; Model_2 = arma(Y, order=c(3,0,3))
sum(Y^2)
[1] 3684.08

Model_1
Call: arma(x = Y, order = c(2, 0, 3))
Coefficients:
      ar1      ar2      ma1      ma2      ma3  intercept
    0.4036  0.4609  0.1826  0.0271  0.6176   -0.4641
s.e.    0.0636  0.0631  0.0513  0.0477  0.0404    0.6786
sigma^2 estimated as 1.081: log likelihood = -592.26, aic = ?

Model_2
Call: arma(x = Y, order = c(3, 0, 3))
Coefficients:
      ar1      ar2      ar3      ma1      ma2      ma3  intercept
    0.3237  0.4004  0.1455  0.2359  0.0986  0.5868   -0.4609
s.e.    0.0797  0.0731  0.0771  0.0652  0.0578  0.0475    0.7346
sigma^2 estimated as 1.072: log likelihood = -590.51, aic = ?
```

Attempt

6.3 Model Diagnostics

Recall if $\{Y_t\} \sim \text{ARIMA}(p, d, q)$, we have $\phi(B)(1 - B)^d Y_t = \theta(B)Z_t$, where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. However, if the fitted model is inappropriate, the fitted residual $\{\hat{Z}_t\}$ may NOT be a white noise sequence. Therefore, it is natural to test whether the fitted model is appropriate by testing whether the estimated residuals are a white noise sequence. Recall in tutorial 02,

$$\text{Under } H_0 : \rho(k) = 0, \quad \sqrt{n}r_k \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty.$$

Those quantities can help to test the existence of significant dependence. However, those tests only focused on a single lag size k . It motivates us to study more general hypotheses and associated tests.

Ljung-Box Test (Portmanteau Statistics)

Theorem 1. Let $r_Z(j)$ be the sample ACF of $\{\hat{Z}_t\}$ and ρ_Z be the ACF of the true noise sequence.

$$H_0 : \rho_Z(k) = 0 \text{ whenever } |k| \leq h \quad \text{against} \quad H_1 : \rho_Z(k) \neq 0 \text{ for some } |k| \leq h$$

for some prespecified h , The **Ljung-Box Test** is defined by

$$Q(h) = n(n+2) \sum_{j=1}^h \frac{\hat{r}_Z^2(j)}{n-j}$$

and $Q(h) \xrightarrow{d} \chi^2(h-p-q)$ under H_0 as $n \rightarrow \infty$.

Remark 4. There are several remarks about the Ljung-Box test

1. A common choice of h lies between 10 and 30.
2. If $Q(h) \geq \chi_{h-p-q, 0.95}^2$, H_0 is rejected.
3. If H_0 is not rejected, then the model is not a bad fit to the data.

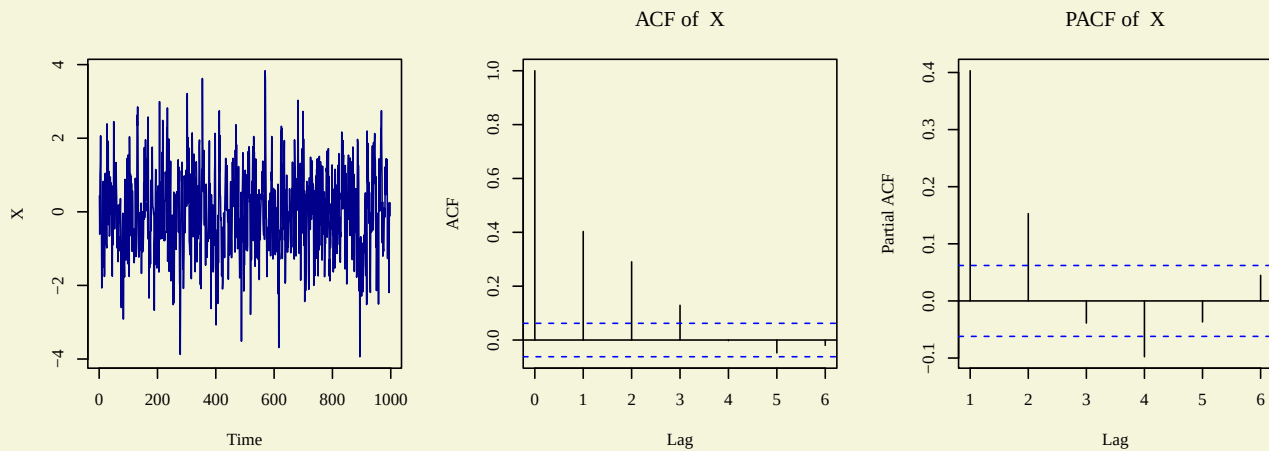
(★★☆) Model Diagnostics

Exercise 3. After we have fitted $Y_1, \dots, Y_{100}$ to the ARMA(1, 2) model, we have the following information regarding the ACF of the estimated residual: $\hat{r}_Z(1) = -0.1$, $\hat{r}_Z(2) = 0.2$ and $|\hat{r}_Z(k)| < 0.05$ whenever $k \geq 3$. Perform the Ljung-box test with $h = 10$ to test whether the fitted model is a good fit.

Solution

(★★☆) Comprehensive Exercise

 **Exercise 4.** Given the time series $Y_1, \dots, Y_n$ ($n = 999$) and $X_t = Y_t - 0.6Y_{t-1} - 0.3Y_{t-2}$, the time-series plot, ACF and PACF plot of $\{X_t\}$ is given in the following figure

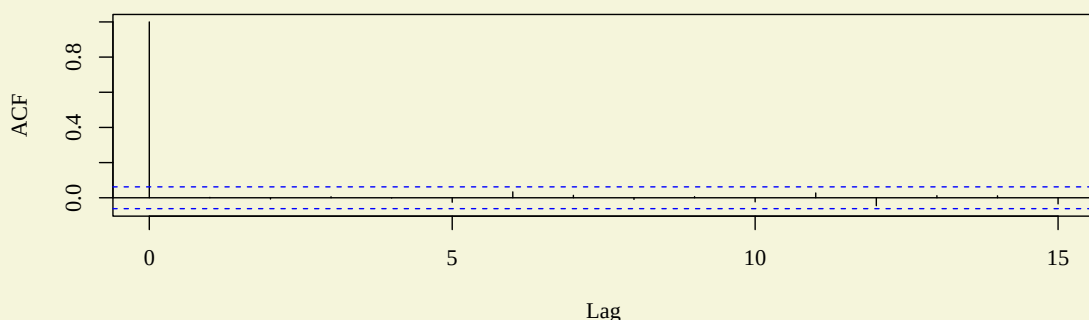


- (a) By observing the above graphs, suggest the most suitable AR or MA model for $\{X_t\}$. Hence, suggest a suitable ARMA(p, q) model for modeling $\{Y_t\}$.
- (b) Kevin suggests that ARMA(2,2) is a more suitable model for prediction based on AIC. Do you agree with him according to the following R-output? (p, q is the order of the ARMA model in (a))

```
Call: arima(x = Y, order = c(p, 0, q)) # Model 1
Coefficients:
      ar1      ar2      ma1      ma2      ma3  intercept
      1.0007  -0.0943  -0.0621  0.3203  0.0726   -0.2451
s.e.    0.2867    0.2629    0.2857  0.0316  0.0953    0.4565

sigma^2 estimated as 1.05: log likelihood = -1442.93
Call: arima(x = Y, order = c(2, 0, 2)) # Model 2
Coefficients:
      ar1      ar2      ma1      ma2  intercept
      1.1996  -0.2752  -0.2627  0.3206   -0.2421
s.e.    0.0887    0.0846    0.0839  0.0330    0.4501
sigma^2 estimated as 1.05: log likelihood = -1443.17
```

- (c) Jensen claimed that as $\hat{\sigma}_1^2 = \hat{\sigma}_2^2$ and those two models have the same order in the AR-polynomial, hence they are indifferent if we choose the model based on FPE. Do you agree with him?
- (d) Check the goodness of fit of the chosen model in part (b) by finding an upper bound for the Portmanteau test with $h = 15$ and refer to the following ACF plot of the estimated residual. Notice that the blue line is the 95% CI for ACF. (Remark: $\chi_{15,0.975}^2 = 27.49$ and $\chi_{11,0.975}^2 = 21.92$)



Attempt

6.4 R-programming

This section provides some basic techniques and common R-command used in time-series analysis.

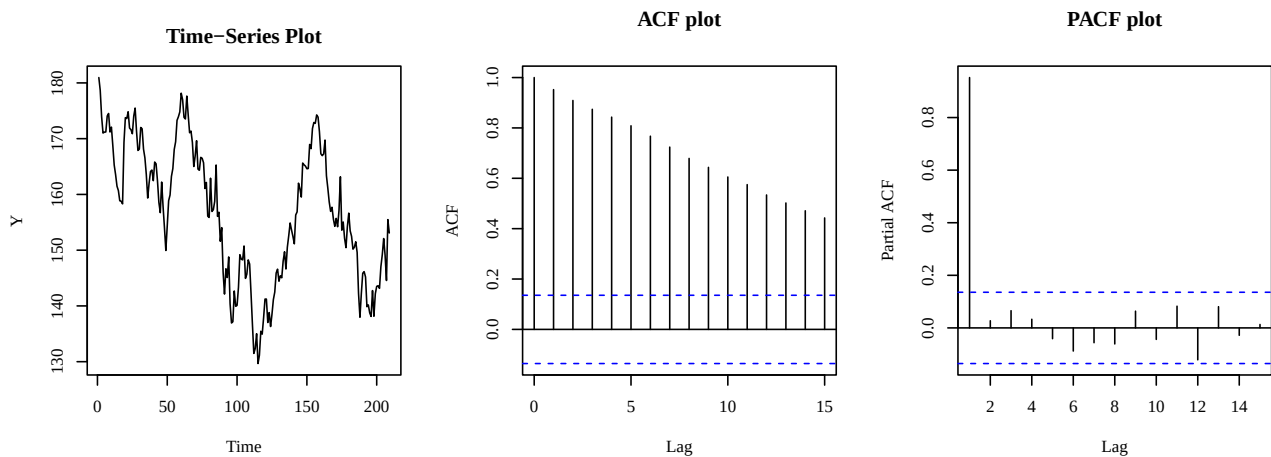
1. Load the time-series data. Let us take the stock price of Apple as an example.

```
install.packages("tseries")
library("tseries")
X0 = get.hist.quote(instrument="AAPL", start="2022-01-01", end="2022-11-01",
  quote="Adjusted", provider=c("yahoo"), compression="d", retclass=c("ts"))
X = X0[-which(is.na(X0)==1)] # Exclude data of non-trading days
```

Remark 5. Indeed, we should do the modeling for the log-return instead of the stock price. But let us do it in a simple way here for illustration purpose.

2. We first visualize the data through the time-series plot, ACF plot, and PACF plot.

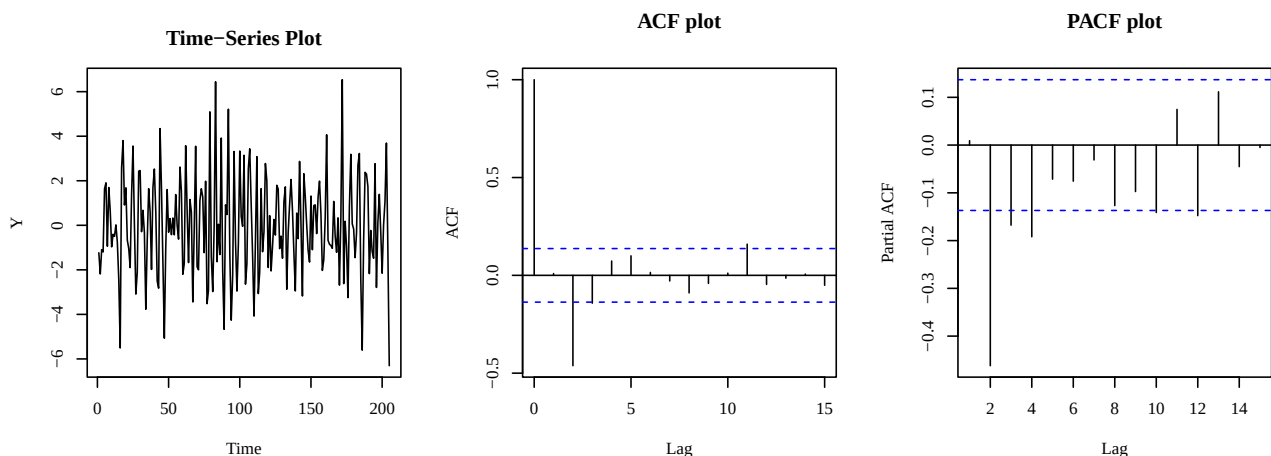
```
par(mfrow=c(1,3))
ts.plot(X,main="Time-Series Plot")
acf(X,lag.max=15,main="ACF plot")
pacf(X,lag.max=15,main="PACF plot")
```



From the plot, it is obvious that the data is highly non-stationary.

3. Apply the technique in Tutorial 01 to get a sequence of estimated noise. Suppose that there is no seasonal effect. Consider filter $(0.2, 0.2, 0.2, 0.2, 0.2)$, define $\hat{T}_t = \sum_{r=-2}^2 a_r X_{t+r}$ and let $Y_t = X_t - \hat{T}_t$.

```
n = length(X) ; That = rep(NA,n-4)
for(i in 1:(n-4)){That[i] = mean(X[i:(i+4)])}
Y = X[3:(n-2)] - That ; n = length(Y)
ts.plot(Y,main="Time-Series Plot")
acf(Y,lag.max=15,main="ACF plot") ; pacf(Y,lag.max=15,main="PACF plot")
```



From the plot, the transformed series $\{Y_t\}$ is more likely to be stationary than $\{X_t\}$.

4. We then conduct the model fitting (By Method of Moment or Leas-Square method or MLE).

Method of Moment Estimator for MA model (With R)

 **Exercise 5.** Suppose $\{Y_t\} \sim \text{MA}(2)$, i.e., $Y_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$ with $\{Z_t\} \sim \text{WN}(0, 1)$. Derive a possible MM estimator for θ_1 and θ_2 . Evaluate them through R.

Solution

$\{Y_t\} \sim \text{MA}(2)$ is causal and therefore stationary. We have

$$\gamma(k) = (1 + \theta_1^2 + \theta_2^2)\mathbf{1}(k=0) + \theta_1(1 + \theta_2)\mathbf{1}(|k|=1) + \theta_2\mathbf{1}(|k|=2)$$

$$\bullet \gamma(2) = \theta_2 \Rightarrow \hat{\theta}_2 = C_2. \quad \bullet \gamma(1) = \theta_1(1 + \theta_2) \Rightarrow \hat{\theta}_1 = C_1/(1 + C_2)$$

By the R-code below, we have $\hat{\theta}_1 = -0.03585743$ and $\hat{\theta}_2 = -2.22720951$.

```
# Extract value of sample ACVF
C = as.numeric(acf(Y,type="covariance",plot=FALSE)$acf)
(thetaMM = c(C[2]/(1+C[3]), C[3])) # Value of thetahat1 and thetahat2
-0.03585743 -2.22720951
```

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Least-Square Estimator for AR model (With R)

 **Exercise 6.** Suppose $\{Y_t\} \sim \text{AR}(3)$, i.e., $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3} + Z_t$ with $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. Evaluate the Least-square estimator of ϕ_1 , ϕ_2 , ϕ_3 and σ^2 . Find 95% CI for ϕ_1 , ϕ_2 and $\phi_2 + \phi_3$.

Solution

It is equivalent to regressing Y_t against covariates Y_{t-1} , Y_{t-2} and Y_{t-3} (without intercept term).

Also, recall from Tutorial 5 that $\sqrt{n}(\hat{\phi} - \phi) \xrightarrow{d} N_3(0, \sigma^2 \Gamma_3^{-1})$, where $\Gamma_3(i, j) = \gamma(|i - j|)$. Notice that for $\Sigma := \sigma^2 \Gamma_3^{-1}$, we have

$$\lim_{n \rightarrow \infty} n \text{Var}(\hat{\phi}_2 + \hat{\phi}_3) = \lim_{n \rightarrow \infty} n \text{Var}(\hat{\phi}_2) + \lim_{n \rightarrow \infty} n \text{Var}(\hat{\phi}_3) + 2 \lim_{n \rightarrow \infty} n \text{Cov}(\hat{\phi}_2, \hat{\phi}_3) = \Sigma(2, 2) + \Sigma(3, 3) + 2\Sigma(2, 3).$$

By the R-code below, we have $\hat{\phi}_1 = -0.08$, $\hat{\phi}_2 = -0.482$, $\hat{\phi}_3 = -0.184$.

```
LSE_fit = lm(Y[4:n] ~ Y[3:(n-1)] + Y[2:(n-2)] + Y[1:(n-3)] - 1)
(phi = as.numeric(LSE_fit$coefficients))
[1] -0.08034209 -0.48242400 -0.18428814
(sigma2 = sum((LSE_fit$residuals)^2)/(n-2*3))
[1] 3.693695
C = as.numeric(acf(Y, type="covariance", plot=FALSE)$acf)
Gamma_p = array(NA,dim=c(3,3))
for(i in 1:3){for(j in 1:3) Gamma_p[i,j] = C[abs(i-j)+1]}
Sigma_hat = sigma2*solve(Gamma_p) # Estimated Asymptotic Covariance Matrix
V1 = qnorm(0.975)*sqrt(Sigma_hat[1,1]/n)
V2 = qnorm(0.975)*sqrt(Sigma_hat[2,2]/n)
V23 = qnorm(0.975)*sqrt((Sigma_hat[2,2]+Sigma_hat[3,3]+2*Sigma_hat[2,3])/n)
(CI1 = c(phi[1]-V1,phi[1]+V1)) # CI for phi1
[1] -0.21279810 0.05211397
(CI2 = c(phi[2]-V2,phi[2]+V2)) # CI for phi2
[1] -0.5999146 -0.3649336
(CI23 = c(phi[2]+phi[3]-V23,phi[2]+phi[3]+V23)) # CI for phi1+phi2
[1] -0.8424410 -0.4909837
```

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Yule-Walker Estimator for AR model (With R)

 **Exercise 7.** Suppose $\{Y_t\} \sim \text{AR}(4)$, i.e., $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3} + \phi_4 Y_{t-4} + Z_t$ with $\{Z_t\} \sim \text{WN}(0, \sigma^2)$. Evaluate the Yule-Walker estimator of ϕ_1, ϕ_2, ϕ_3 and ϕ_4 .

Solution

Recall from Tutorial 5, $\hat{\phi} = M^{-1}(r_1, \dots, r_4)^T$, where $M(i, j) = r_{|i-j|}$. From the R-code below, we have $\hat{\phi} = (-0.096, -0.548, -0.180, -0.192)$.

```
r = as.numeric(acf(Y, type="correlation", plot=FALSE)$acf)[1:5] #Compute r0,...,r4
M = array(NA, dim=c(4,4))
for(i in 1:4){for(j in 1:4) M[i,j] = r[abs(i-j)+1]}
(phi_YW = solve(M)%*%r[2:5]) # Yule-Walker Estimator
      [,1]
[1,] -0.0962834
[2,] -0.5479180
[3,] -0.1799613
[4,] -0.1919126
```

Conditional Least Square Estimator for ARMA model (With R)

 **Exercise 8.** Assume that $\{Y_t\} \sim \text{ARMA}(2, 3)$, i.e., $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2} - \theta_3 Z_{t-3}$. Find the CLS estimator for $\phi_1, \phi_2, \phi_3, \theta_1, \theta_2$ and σ^2 .

Solution

Write $Z_t = Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \theta_3 Z_{t-3}$ and define $Z_s = Y_s = 0$ for $s \leq 0$. After we obtain the CLS estimate, we can define $\hat{Z}_t = Y_t - \hat{\phi}_1 Y_{t-1} - \hat{\phi}_2 Y_{t-2} + \hat{\theta}_1 \hat{Z}_{t-1} + \hat{\theta}_2 \hat{Z}_{t-2} + \hat{\theta}_3 \hat{Z}_{t-3}$ and hence $\hat{\sigma}^2 = \sum_{t=1}^n \hat{Z}_t^2 / n$.

```
S = function(beta){
  phi1 = beta[1] ; phi2 = beta[2] ; theta1 = beta[3] ; theta2 = beta[4] ; theta3 =
    beta[5]
  Z = rep(NA, n)
  Z[1] = Y[1] ; Z[2] = Y[2] - phi1*Y[1] + theta1*Z[1]
  Z[3] = Y[3] - phi1*Y[2] - phi2*Y[1] + theta1*Z[2] + theta2*Z[1]
  for(k in 4:n){
    Z[k] = Y[k] - phi1*Y[(k-1)] - phi2*Y[(k-2)] + theta1*Z[(k-1)] + theta2*Z[(k-2)]
      + theta3*Z[(k-3)]
  }
  return(sum(Z^2))
}
CLS_fit = optim(c(0.1,0.1,0.1,0.1,0.1), S) # Optimize the function S wrt beta
(phiihat = CLS_fit$par[1:2])
[1] 0.2207534 -0.2051981
(thetahat = CLS_fit$par[3:5])
[1] 0.39629000 0.39611940 0.03076929
(sig2hat = CLS_fit$value/n)
[1] 3.317136
```

Maximum Likelihood Estimator for ARMA model (With R)

 **Exercise 9.** Assume that $\{Y_t\} \sim \text{ARMA}(2, 3)$, i.e., $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2} - \theta_3 Z_{t-3}$. Find the MLE for $\phi_1, \phi_2, \phi_3, \theta_1, \theta_2$ and σ^2 and also the value of the maximized log-likelihood.

Solution

$\hat{\phi}_1 = 0.130, \hat{\phi}_2 = -0.138, \hat{\theta}_1 = 0.378, \hat{\theta}_2 = 0.529, \hat{\theta}_3 = -0.093$ and $\hat{\sigma}^2 = 3.11$.

```
MLE_fit = arima(Y, order=c(2,0,3), include.mean = FALSE)
MLE_fit$coef
      ar1      ar2      ma1      ma2      ma3
0.12987568 -0.13766270 -0.37820460 -0.52862527 -0.09316591
MLE_fit$loglik
[1] -409.6758
MLE_fit$sigma2
[1] 3.110787
```

 **Remark 6.** In R, the ARMA model is of the form $Y_t = \sum_{k=1}^p \phi_k Y_{t-k} + Z_t + \sum_{k=1}^q \theta_k Z_{t-k}$ instead of $Y_t = \sum_{k=1}^p \phi_k Y_{t-k} + Z_t - \sum_{k=1}^q \theta_k Z_{t-k}$, i.e., we have to be careful with the sign of the MA coefficient.

5. After fitting the model, say the $\text{ARMA}(p_1, q_1)$ model. We can compare its performance with the $\text{ARMA}(p_2, q_2)$ model with different criteria.

Model Selection

 **Exercise 10.** Consider the class of $\text{ARMA}(p, q)$ model with $1 \leq p, q \leq 5$. Which model is the best in terms of AIC, AICC, and BIC?

Solution

According to the R-code below, both AIC and AICC suggest $\text{ARMA}(5, 4)$ while BIC suggest $\text{ARMA}(4, 5)$.

```
IC=function(x, order.input){
  fit = arima(x, order=c(order.input[1], 0, order.input[2]));
  n = length(x) ; p = order.input[1] ; q = order.input[2] ; sig2 = fit$sigma2
  AIC = fit$aic
  AICC = AIC - 2*(p+q+1) + 2*(p+q+1)*n/(n-p-q-2)
  BIC=(n-p-q)*log(n*sig2/(n-p-1))+n*(1+log(sqrt(2*pi)))+
    (p+q)*log((sum(x^2)-n*sig2)/(p+q))
  FPE = sig2*(n+p)/(n-p) # Only valid for AR model
  out = t(as.matrix(c(AIC, AICC, BIC, FPE))); colnames(out)=c("AIC", "AICC", "BIC", "FPE")
  return(out)
}

out = array(NA, dim=c(5,5,4))
for(i in 1:5){for(j in 1:5) out[i,j,] = IC(Y, c(i,j)) }
(order_AIC = arrayInd(which.min(out[, , 1]), c(5,5)))
  [,1] [,2]
[1,]    5    4
(order_AICC = arrayInd(which.min(out[, , 2]), c(5,5)))
  [,1] [,2]
[1,]    5    4
(order_BIC = arrayInd(which.min(out[, , 3]), c(5,5)))
  [,1] [,2]
[1,]    4    5
```

6. After we have chosen a model through model selection, we should conduct residual analysis, i.e., consider the time-series plot, ACF, and PACF plot, and carry out the Portmanteau test.

Residual Analysis

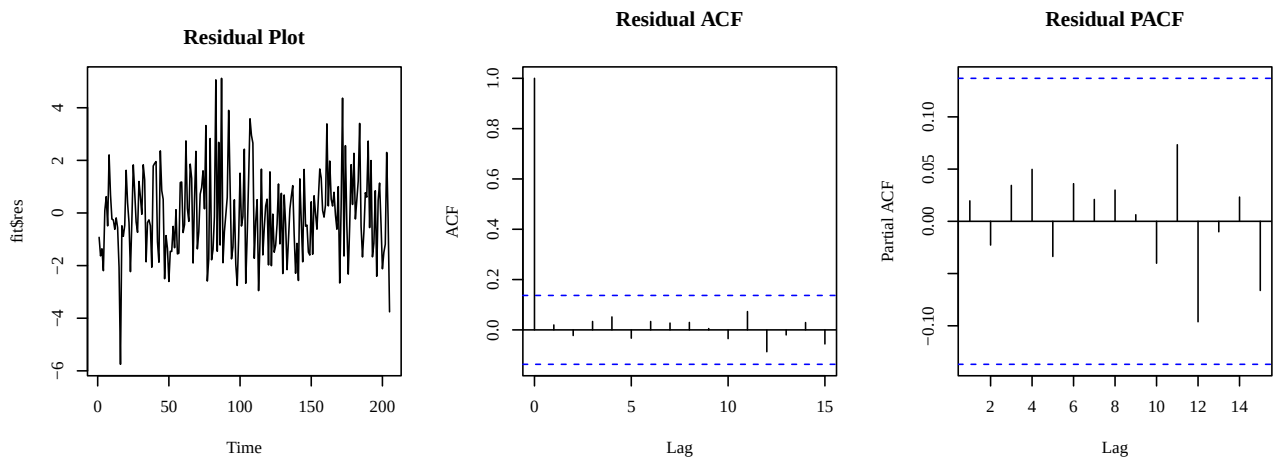
 **Exercise 11.** Suppose we choose the ARMA(5,4) model for $\{Y_t\}$. We want to check its goodness of fit through the time-series plot, ACF, and PACF plot. Also, perform the portmanteau test with $h = 15$. Clearly state the hypothesis and your conclusion.

Solution

Let Z be the noise under this ARMA(5,4) model and $\rho_Z(\cdot)$ as ACF of Z , we are testing

$$H_0 : \rho_Z(k) = 0 \text{ whenever } |k| \leq 15 \quad \text{against} \quad H_1 : \rho_Z(k) \neq 0 \text{ for some } |k| \leq 15$$

According to the result in the R-code below, we do NOT reject H_0 , i.e. at least ARMA(5,4) is NOT a bad fit for $\{Y_t\}$. Also, from the graph below, we can see that both ACF and PACF of residuals are not significant. Suggesting that the residual might be a white-noise sequence.



```
p = 5 ; q = 4 ; h = 15
fit = arima(Y,order=c(p,0,q))
par(mfrow=c(1,3))
ts.plot(fit$res,main="Residual Plot")
r.z = as.numeric(acf(fit$res,lag.max=h,main="Residual ACF")$ acf)
pacf(fit$res,lag.max=h,main="Residual PACF")
Qh = n*(n+2)*sum((r.z[-1]^2)/(n-(1:h)))
Qh > qchisq(0.95,h-p-q)
[1] FALSE
```

1
2
3
4
5
6
7
8
9